Advanced Trigonometry

Section 1: Angle Sum, Difference & Double-Angle Formulas

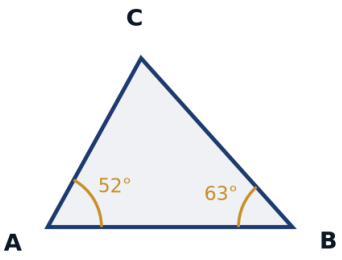
1 If $\sin\theta = \frac{3}{5}$ and θ is in Quadrant II, what is the exact value of $\sin(2\theta)$?

a. $-\frac{24}{25}$

b. $\frac{24}{25}$

c. $-\frac{7}{25}$

d. $\frac{7}{25}$



2 What is the exact value of $\cos(75^\circ)$ using the angle sum formula $\cos(45^\circ + 30^\circ)$?

a. $\frac{\sqrt{6}-\sqrt{2}}{4}$

b. $\frac{\sqrt{6}+\sqrt{2}}{4}$

c. $\frac{\sqrt{2}-\sqrt{6}}{4}$

d. $\frac{\sqrt{3}-1}{2}$

3 Which of the following is equivalent to $\frac{1-\cos(2x)}{2}$?

a. $\sin^2 x$

b. $\cos^2 x$

c. $\tan^2 x$

d. $\sin(2x)$

Section 2: Solving Trigonometric Equations

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4 Solve the equation $2\sin^2 x - 3\sin x + 1 = 0$ for all solutions in the interval $[0, 2\pi)$.

a. $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{\pi}{2}$

b. $\frac{\pi}{3}, \frac{2\pi}{3}, \pi$

c. $\frac{\pi}{6}, \frac{7\pi}{6}$

d. $\frac{\pi}{2}, \frac{3\pi}{2}$

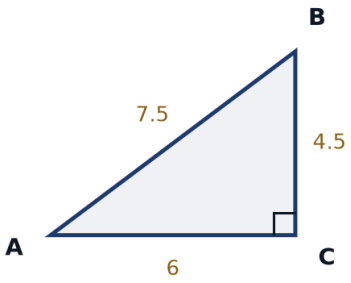
5 How many solutions does the equation $\cos(2x) = 0$ have in the closed interval $[0, 2\pi]$?

a. 2

b. 4

c. 6

d. 8



6 Solve for $x \in [0, \pi]$: $\sqrt{3}\tan(2x) = 1$.

a. $\frac{\pi}{12}, \frac{7\pi}{12}$

b. $\frac{\pi}{6}, \frac{5\pi}{6}$

c. $\frac{\pi}{3}$

d. $\frac{\pi}{4}, \frac{3\pi}{4}$

Section 3: Inverse Trigonometric Functions

7 What is the exact value of $\sin(\arccos(-\frac{1}{2}))$?

a. $\frac{\sqrt{3}}{2}$

b. $-\frac{\sqrt{3}}{2}$

c. $\frac{1}{2}$

d. $-\frac{1}{2}$

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8 Evaluate $\tan(\arcsin(\frac{5}{13}))$.

a. $\frac{5}{12}$

b. $\frac{12}{5}$

c. $\frac{5}{13}$

d. $\frac{12}{13}$

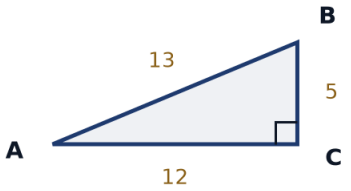
9 What is the range of the principal inverse tangent function $f(x) = \arctan(x)$?

a. $(-\frac{\pi}{2}, \frac{\pi}{2})$

b. $[-\frac{\pi}{2}, \frac{\pi}{2}]$

c. $[0, \pi]$

d. $(0, \pi)$



Section 4: Law of Tangents & Polar Trigonometry

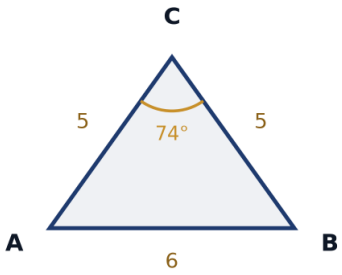
10 Which identity represents $\cos(A - B)$?

a. $\cos A \cos B + \sin A \sin B$

b. $\cos A \cos B - \sin A \sin B$

c. $\sin A \cos B - \cos A \sin B$

d. $\sin A \cos B + \cos A \sin B$



11 In $\triangle ABC$, $a = 7$, $b = 5$, and $c = 8$. What is $\cos B$?

a. $\frac{11}{14}$

b. $\frac{1}{2}$

c. $\frac{5}{7}$

d. $\frac{13}{16}$

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12 What is the maximum value of the function $f(x) = 3\sin x - 4\cos x$?

a. 1

b. 5

c. 7

d. $\sqrt{7}$

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